Signal Processing: Methods and Algorithms
Laboratory 6

1 Spectrogram

Write a Matlab function that computes the spectrogram. Modify the code developed for the sliding estimators. The function must have the following syntax:

`[P] = spectrogram(x,Istep,Nfft,h,PlotFlag)`

where:

1. `x` is a row vector with `N` elements, which represents the signal to be transformed.

2. `Istep` is the time step used in the evaluation of the spectrogram. The spectrogram will be evaluated at the discrete time instants `I=1:Istep:N`.

3. `Nfft` is the number of points for the FFT algorithm. Remember that the command

$$\mathtt{X} = \mathtt{fft}(\mathtt{x}, \mathtt{Nfft}) \tag{1}$$

computes the FFT of `x` and generates an output vector `X` with `Nfft` elements (a zero padding technique is used when `Nfft` > `N`).

4. `h` is the window, a row vector with `Nw` elements (assume `Nw` to be even). The command

$$\mathtt{h} = \mathtt{ones}(1, \mathtt{Nw}) \tag{2}$$

generates a rectangular window. Inside the Spectrogram function, the length `Nw` of the window `h` can be obtained with the command

$$\mathtt{Nw} = \mathtt{length}(\mathtt{h}) \tag{3}$$

Zero-pad the signal by adding `Nw/2` zeros at the beginning and `Nw/2-1` zeros at the end of it. The following code can be used to generate the truncated signal `xn` (where `n` spans the time values `0:Istep:N-1`)

```
n1=n-Nw/2;
n2=n+Nw/2-1;
if n1<0
    xn=[zeros(1,abs(n1)) x(1:n2+1)];
elseif n2>N-1
    xn=[x(n1+1:N) zeros(1,n2-(N-1))];
else
    xn=x(n1+1:n2+1);
end
```

5. `PlotFlag` is a flag that controls the graphic output of the function. When `PlotFlag=1` the spectrogram `P` is displayed in a Matlab figure. To display the spectrogram use the following command

$$\mathtt{figure}, \mathtt{imagesc}(\mathtt{I}, \mathtt{f}, \mathtt{P}), \mathtt{axis\ xy}; \tag{4}$$

where `I=1:Istep:N` is the vector of the discrete time instants at which the spectrogram is evaluated, while `f` is the frequency axis, a vector of `Nfft/2+1` elements going from $f = 0$ to $f = 1/2$.

6. `P` is the matrix containing the computed spectrogram. `P` has `Nfft/2+1` rows, corresponding to the nonnegative frequencies of the spectrum, evaluated with the FFT algorithm (the first `Nfft/2+1` elements of the output vector of the function FFT). The number of columns of `P` equals the number of elements of `I`.

2 Time-frequency analysis of nonstationary signals

This exercise deals with the generation of a series of signals whose frequency content changes with time (except for the first case). The generated signals must then be analyzed with the spectrogram function described in the previous section.

2.1 Simple sinusoid

Consider the sinusoid

$$x[n] = \cos 2\pi f_0 n \tag{5}$$

where $f_0 = 0.15$. Do the following:

1. Generate and plot $N = 1000$ sample of $x[n]$.
2. Compute and plot the spectrogram P_x of $x[n]$.

2.2 Four sinusoids

Consider a signal $x[n]$ made by four short duration sinusoids

$$x[n] = \begin{cases} \cos 2\pi f_1 n, & 0 \le n \le 249 \\ \cos 2\pi f_2 n, & 250 \le n \le 499 \\ \cos 2\pi f_3 n, & 500 \le n \le 749 \\ \cos 2\pi f_4 n, & 750 \le n \le 999 \end{cases} \tag{6}$$

where $f_1 = 0.1$, $f_2 = 0.2$, $f_3 = 0.3$, and $f_4 = 0.4$. Do the following:

1. Generate and plot $N = 1000$ samples of $x[n]$.
2. Compute and plot the spectrogram P_x of $x[n]$.
3. Change the length N_w of the window and study its effect on the time localization and frequency resolution.

2.3 Linear chirp

Generate $N = 10000$ samples of a linear chirp, defined in discrete time by

$$x[n] = \cos(2\pi f_0 n + \beta n^2/2) \tag{7}$$

Choose f_0 and β so that the instantaneous frequency $f_i[n]$ satisfies

$$f_i[0] = f_1 \tag{8}$$
$$f_i[N] = f_2 \tag{9}$$

where $f_1 = .1$ and $f_2 = .3$. **Hint:** the parameters f_0 and β are given by

$$f_0 = f_1 \tag{10}$$
$$\beta = 2\pi \frac{f_2 - f_1}{N} \tag{11}$$

Do the following:

1. Generate and plot $x[n]$.
2. Compute and plot the spectrogram P_x of $x[n]$.

2.4 Multi-component signal

The file `TF_Signal.txt` contains a nonstationary signal $x[n]$ made by four components, namely, a quadratic chirp, a short duration sinusoid, a sinusoid, and a bandpass noise. Compute and represent the spectrogram of $x[n]$, then identify the four components.